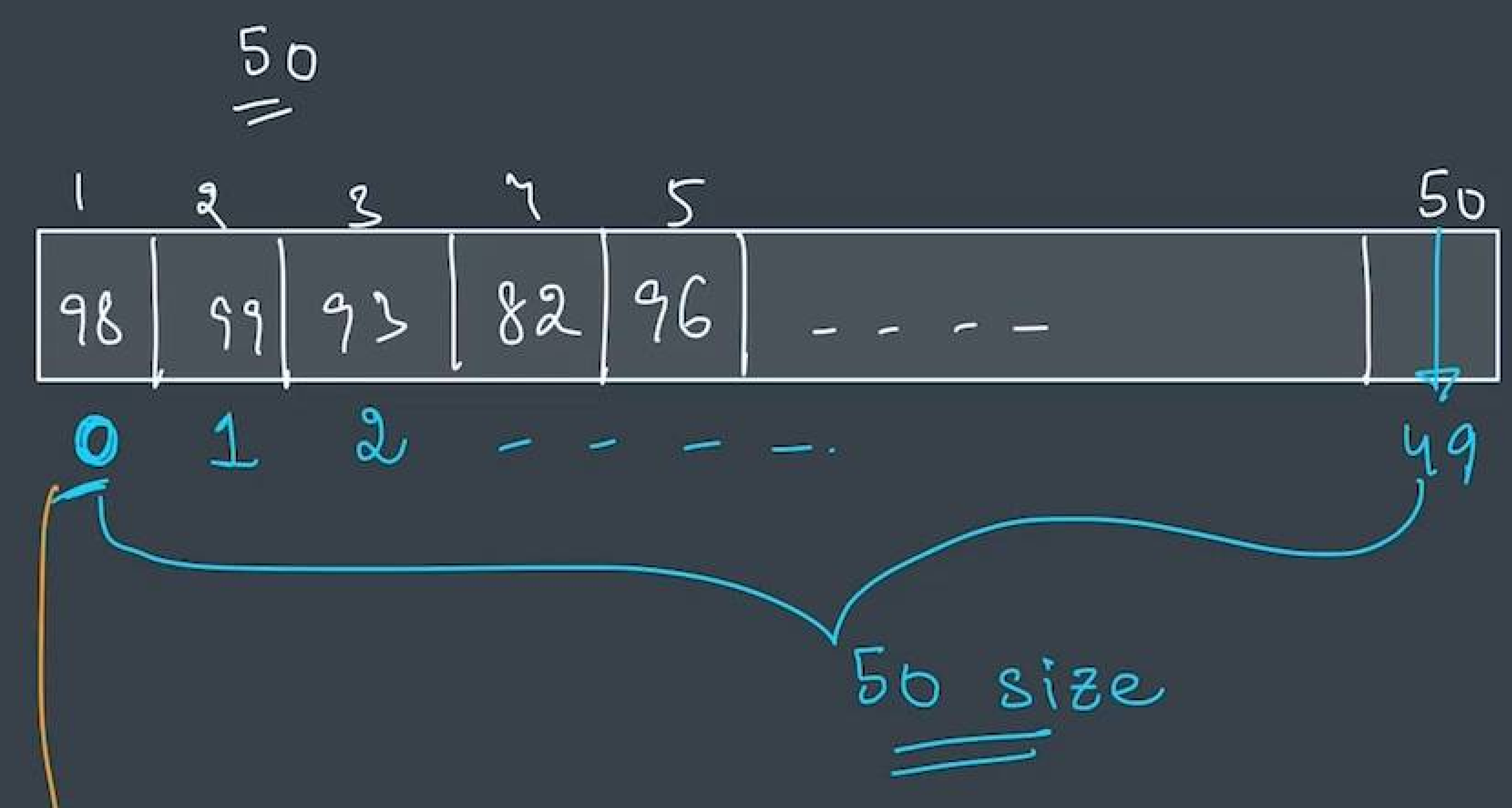
Arrays

int phy = 98

int chem = 99

int math = 93



0-based indexing

C++ ✓

java ✓

Definition of an Array

List of Elements of the same type placed in a contiguous memory location

"apple" "mango" 4.9 8

X

1, 2, 3, 4, 5, 6 → int

"apple", "mango" → string

1.1, 4.9, 8.6, 2.8 → float

Definition of an Array

List of Elements of the same type placed in a contiguous memory location

"apple" "mango" 4.9 8

X

1, 2, 3, 4, 5, 6 → int

"apple", "mango" → string

1.1, 4.9, 8.6, 2.8 → float

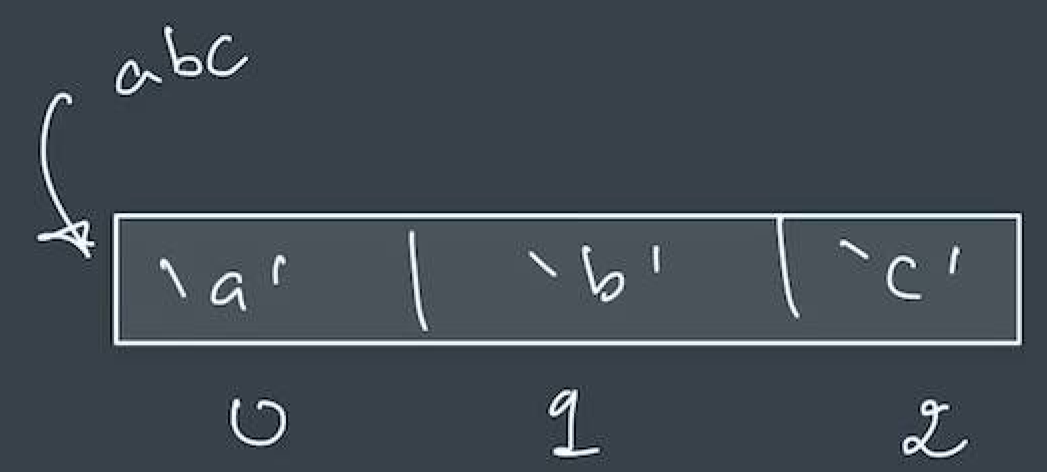
'a', 'b', 'c' → ch

Definition of an Array

List of Elements of the **same** type
placed in a **contiguous** memory location

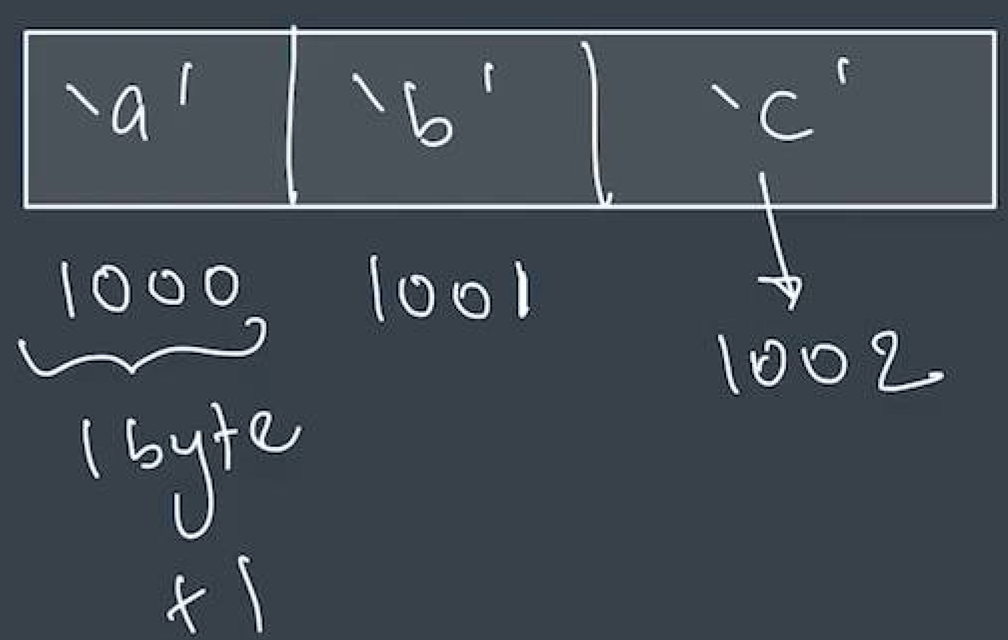
Here char size is taken as 1 byte for simplicity. Java char takes up 2 bytes in memory.

ARRAYS



Definition of an Array

List of Elements of the **same** type placed in a **contiguous** memory location



Operations in Arrays

- Create
- Input
- Output
- Update

Creating an Array

dataType arrayName[ ] = new dataType[size];

```java
//creating an array

int marks[] = new int[50];

int numbers[] = {1, 2, 3};

int moreNumbers[] = {4, 5, 6};

String fruits[] = {"apple", "mango", "orange"};
```

www.apnacollege.in – To exit full screen, press Esc

ArraysCC.java U

ArraysCC.java

```java
import java.util.*;

public class ArraysCC {
    public static void main(String args[]) {
        int marks[] = new int[50];
    }
}
```


02:49 / 04:59

ArraysCC.java U

ArraysCC.java

```java
import java.util.*;

public class ArraysCC {
    public static void main(String args[]) {
        int marks[] = new int[50];

        int numbers[] = {1, 2, 3}; // 3
    }
}
```


ArraysCC.java U

ArraysCC.java

```java
import java.util.*;

public class ArraysCC {
    public static void main(String args[]) {
        int marks[] = new int[100];

        Scanner sc = new Scanner(System.in);
        // int phy;
        // phy = sc.nextInt();

        marks[0] = sc.nextInt(); //phy
        marks[1] = sc.nextInt(); //chem
        marks[2] = sc.nextInt(); // math

        System.out.println("phy : " + marks[0]);
        System.out.println("chem : " + marks[1]);
        System.out.println("math : " + marks[2]);
    }
}
```

ArraysCC.java U ×

ArraysCC.java

```java
    public static void main(String args[]) {
        int marks[] = new int[100];

        Scanner sc = new Scanner(System.in);
        // int phy;
        // phy = sc.nextInt();

        marks[0] = sc.nextInt(); //phy
        marks[1] = sc.nextInt(); //chem
        marks[2] = sc.nextInt(); // math

        System.out.println("phy : " + marks[0]);
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE

1: zsh

```
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
97
99
95
phy : 97
chem : 99
math : 95
shradhakhapra@Shradhas-Air Classroom Codes % 
```

armankhan082020@gmail.com

ArraysCC.java U

ArraysCC.java

```java
    public static void main(String args[]) {
        int marks[] = new int[100];

        Scanner sc = new Scanner(System.in);
        // int phy;
        // phy = sc.nextInt();

        marks[0] = sc.nextInt(); //phy
        marks[1] = sc.nextInt(); //chem
        marks[2] = sc.nextInt(); // math

        System.out.println("phy : " + marks[0]);
        System.out.println("chem : " + marks[1]);
        System.out.println("math : " + marks[2]);

        marks[2] = 100;
        System.out.println("math : " + marks[2]);

    }
}
```

ArraysCC.java U ×

ArraysCC.java

```java
    public static void main(String args[]) {
        int marks[] = new int[100];

        Scanner sc = new Scanner(System.in);
        // int phy;
        // phy = sc.nextInt();

        marks[0] = sc.nextInt(); //phy
        marks[1] = sc.nextInt(); //chem
        marks[2] = sc.nextInt(); // math

        System.out.println("phy : " + marks[0]);
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE 1: zsh

```
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
98
99
95
phy : 98
chem : 99
math : 95
math : 100
shradhakhapra@Shradhas-Air Classroom Codes %
```

ArraysCC.java U

ArraysCC.java

```java
        marks[0] = sc.nextInt(); //phy
        marks[1] = sc.nextInt(); //chem
        marks[2] = sc.nextInt(); // math

        System.out.println("phy : " + marks[0]);
        System.out.println("chem : " + marks[1]);
        System.out.println("math : " + marks[2]);

        marks[2] = marks[2] + 1;
        System.out.println("math : " + marks[2]);

    }
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE 1: zsh

```
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
98
99
95
phy : 98
chem : 99
math : 95
math : 100
shradhakhapra@Shradhas-Air Classroom Codes % 
```

```java
        marks[0] = sc.nextInt(); //phy
        marks[1] = sc.nextInt(); //chem
        marks[2] = sc.nextInt(); // math

        System.out.println("phy : " + marks[0]);
        System.out.println("chem : " + marks[1]);
        System.out.println("math : " + marks[2]);
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE 1: zsh

```
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
98
99
95
phy : 98
chem : 99
math : 95
math : 100
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
^[[A%
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
98
99
95
phy : 98
chem : 99
math : 95
math : 96
shradhakhapra@Shradhas-Air Classroom Codes %
```

main* Run Testcases ⊗ 0 ⚠ 0 Ln 18, Col 1 Spaces: 4 UTF

ArraysCC.java U

ArraysCC.java

```java
public class ArraysCC {
    public static void main(String args[]) {
        int marks[] = new int[100];

        Scanner sc = new Scanner(System.in);
        // int phy;
        // phy = sc.nextInt();

        System.out.println("length of array = " + marks.length);

        // marks[0] = sc.nextInt(); //phy
        // marks[1] = sc.nextInt(); //chem
        // marks[2] = sc.nextInt(); // math

        // System.out.println("phy : " + marks[0]);
        // System.out.println("chem : " + marks[1]);
        // System.out.println("math : " + marks[2]);

        // int percentage = (marks[0] + marks[1] + marks[2]) / 3
        // System.out.println("percentage = " + percentage + "%")
```

ArraysCC.java U ×

ArraysCC.java

```java
public class ArraysCC {
    public static void main(String args[]) {
        int marks[] = new int[100];

        Scanner sc = new Scanner(System.in);
        // int phy;
        // phy = sc.nextInt();
```

PROBLEMS   OUTPUT   TERMINAL   DEBUG CONSOLE   1: zsh

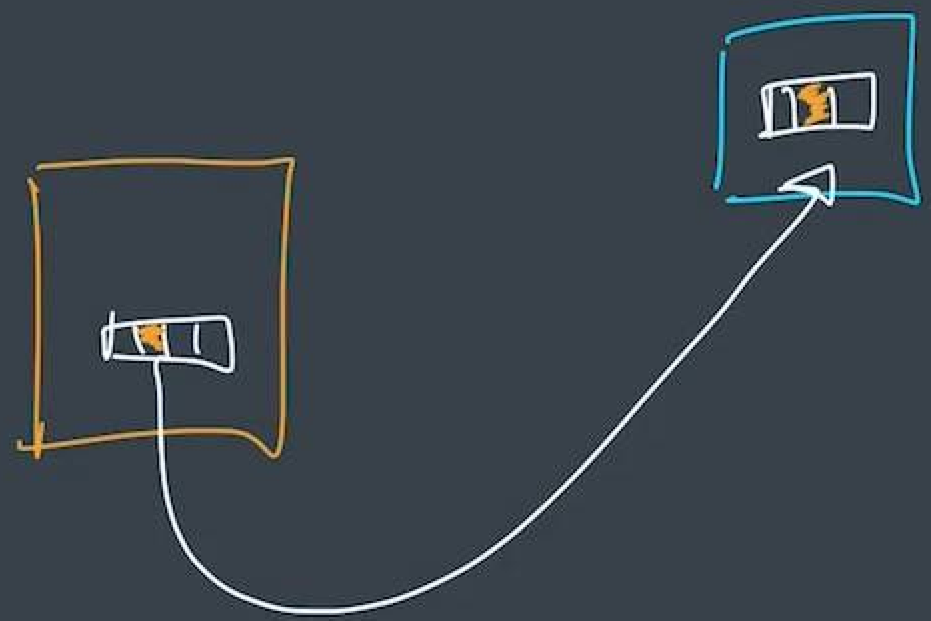
```
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
length of array = 100
shradhakhapra@Shradhas-Air Classroom Codes %
```

Passing arrays as argument

pass by value

main X

by reference



ArraysCC.java U

ArraysCC.java

```java
import java.util.*;

public class ArraysCC {
    public static void update(int marks[]) {
        for(int i=0; i<marks.length; i++) {
            marks[i] = marks[i] + 1;
        }
    }
    public static void main(String args[]) {
        int marks[] = {97, 98, 99};
        update(marks);

        //print our marks
        for(int i=0; i<marks.length; i++) {
            System.out.print(marks[i]+" ");
        }
        System.out.println();
    }
}
```

ArraysCC.java U ×

ArraysCC.java

```java
import java.util.*;

public class ArraysCC {
    public static void update(int marks[]) {
        for(int i=0; i<marks.length; i++) {
            marks[i] = marks[i] + 1;
        }
    }
    public static void main(String args[]) {
        int marks[] = {97, 98, 99};
        update(marks);

        //print our marks
        for(int i=0; i<marks.length; i++) {
            System.out.print(marks[i]+" ");
        }
        System.out.println();
```

PROBLEMS   OUTPUT   TERMINAL   DEBUG CONSOLE   1: zsh

```
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
98 99 100
shradhakhapra@Shradhas-Air Classroom Codes %
```

armankhan082020@gmail.com

ArraysCC.java U ×

ArraysCC.java

```java
public class ArraysCC {
    public static void update(int marks[], int nonChangable) {
        nonChangable = 10;
        for(int i=0; i<marks.length; i++) {
            marks[i] = marks[i] + 1;
        }
    }
    public static void main(String args[]) {
        int marks[] = {97, 98, 99};
        int nonChangable = 5;
        update(marks, nonChangable);
        System.out.println(nonChangable);

        //print our marks
        for(int i=0; i<marks.length; i++) {
            System.out.print(marks[i]+" ");
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE Filter (e.g. text, **/*.ts, !**/nod

No problems have been detected in the workspace.

ArraysCC.java U ×

ArraysCC.java

```java
public class ArraysCC {
    public static void update(int marks[], int nonChangable) {
        nonChangable = 10;
        for(int i=0; i<marks.length; i++) {
            marks[i] = marks[i] + 1;
        }
    }
    public static void main(String args[]) {
        int marks[] = {97, 98, 99};
        int nonChangable = 5;
        update(marks, nonChangable);
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE 1: zsh

```
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
98 99 100
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
5
98 99 100
shradhakhapra@Shradhas-Air Classroom Codes %
```

ArraysCC.java U

ArraysCC.java

```java
        for(int i=0; i<marks.length; i++) {
            marks[i] = marks[i] + 1;
        }
    }
    public static void main(String args[]) {
        int marks[] = {97, 98, 99};
        int nonChangable = 5;
        update(marks, nonChangable);
        System.out.println(nonChangable);

        //print our marks
        for(int i=0; i<marks.length; i++) {
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE 1: zsh

```
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
98 99 100
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
5
98 99 100
shradhakhapra@Shradhas-Air Classroom Codes %
```

Linear Search

find the **index** of element in a given array

key = 10

2	4	6	8	10	12	14	16

Dosa
Chole Bh
Samose
Sandwich
Frooti
Coke

ARRAYS

armankhan082020@gmail.com

ArraysCC.java U ×

ArraysCC.java

```java
import java.util.*;
//Linear Search
public class ArraysCC {
    public static int linearSearch(int numbers[], int key) {

        for(int i=0; i<numbers.length; i++) {
            if(numbers[i] == key) {
                return i;
            }
        }

        return -1;
    }

    public static void main(String args[]) {
        int numbers[] = {2, 4, 6, 8, 10, 12, 14, 16};
        int key = 10;

        int index = linearSearch(numbers, key);
        if(index == -1) {
            System.out.println("NOT found");
        } else {
            System.out.println("key is at index : " + index);
```

ArraysCC.java U

ArraysCC.java

```java
import java.util.*;
//Linear Search
public class ArraysCC {
    public static int linearSearch(int numbers[], int key) {

        for(int i=0; i<numbers.length; i++) {
            if(numbers[i] == key) {
                return i;
            }
        }

        return -1;
    }

    public static void main(String args[]) {
        int numbers[] = {2, 4, 6, 8, 10, 12, 14, 16};
        int key = 10;

        int index = linearSearch(numbers, key);
        if(index == -1) {
            System.out.println("NOT found");
        } else {
            System.out.println("key is at index : " + index);
```

ArraysCC.java U ×

ArraysCC.java

```java
    public static void main(String args[]) {
        int numbers[] = {2, 4, 6, 8, 10, 12, 14, 16};
        int key = 20;

        int index = linearSearch(numbers, key);
        if(index == -1) {
            System.out.println("NOT found");
        } else {
            System.out.println("key is at index : " + index);
        }

    }
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE 1: zsh

```
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
key is at index : 4
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
NOT found
shradhakhapra@Shradhas-Air Classroom Codes % 
```

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ArraysCC.java U

ArraysCC.java

```java
public static void main(String args[]) {
    int numbers[] = {2, 4, 6, 8, 10, 12, 14, 16};
   // String menu[] = {"dosa", "chole bhature", "samosa" .. }
    int key = 20;

    int index = linearSearch(numbers, key);
    if(index == -1) {
        System.out.println("NOT found");
    } else {
        System.out.println("key is at index : " + index);
    }
```

PROBLEMS   OUTPUT   TERMINAL   DEBUG CONSOLE   1: zsh

```
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
key is at index : 4
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
NOT found
shradhakhapra@Shradhas-Air Classroom Codes % 
```

Linear Search

find the **index** of element in a given array

key = 10

2	4	6	8	10	12	14	16

Linear Search

find the index of element in a given array

key = 10

		6	8	10	12	14	16

loop

length

for (i = 0 to n)

worst

n operations

$\boxed{O(n)}$

Largest Number

find the **largest** number in a given array

1	2	6	3	5

Largest Number

largest = $\cancel{-\infty}$ $\cancel{1}$ $\cancel{2}$ 6

6 → largest

find the largest number in a given array

1	2	6	3	5
0	1	2	3	4

ARRAYS

Largest Number

largest = $\cancel{-\infty}$ $\cancel{1}$ $\cancel{2}$ 6

6 → largest

find the largest number in a given array

1	2	6	3	5
0	1	2	3	4

$-\infty$ Integer.MIN_VALUE

$+\infty$ Integer.MAX_VALUE

ARRAYS

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ArraysCC.java 1, U

ArraysCC.java > ArraysCC > main(String[])

```java
import java.util.*;
//Largest Number in Array

public class ArraysCC {
    public static int getLargest(int numbers[]) {
        int largest = Integer.MIN_VALUE; // -infinity

        for(int i=0; i<numbers.length; i++) {
            if(largest < numbers[i]) {
                largest = numbers[i];
            }


        argest;



        oid main(String args[]) {
        [] = {1, 2, 6, 3, 5};
        println("largest value is : " + getLargest(numbers));
```

ArraysCC.java 1, U

ArraysCC.java > ArraysCC > main(String[])

```java
import java.util.*;
//Largest Number in Array

public class ArraysCC {
    public static int getLargest(int numbers[]) {
        int largest = Integer.MIN_VALUE; // -infinity

        for(int i=0; i<numbers.length; i++) {
            if(largest < numbers[i]) {
                largest = numbers[i];
            }
        }

        return largest;
    }

```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
```

ArraysCC.java 1, U

ArraysCC.java > ArraysCC > getLargest(int[])

```java
        int largest = Integer.MIN_VALUE; // -infinity
        int smallest = Integer.MAX_VALUE;

        for(int i=0; i<numbers.length; i++) {
            if(largest < numbers[i]) {
                largest = numbers[i];
            }
            if(smallest > numbers[i]) {
                smallest = numbers[i];
            }
        }
        System.out.println("smallest value is : " + smallest);
        return largest;
    }

```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
largest value is : 6
shradhakhapra@Shradhas-Air Classroom Codes % 
```

Binary Search

key = 10

2	4	6	8	10	12	14

prerequisite - sorted arrays

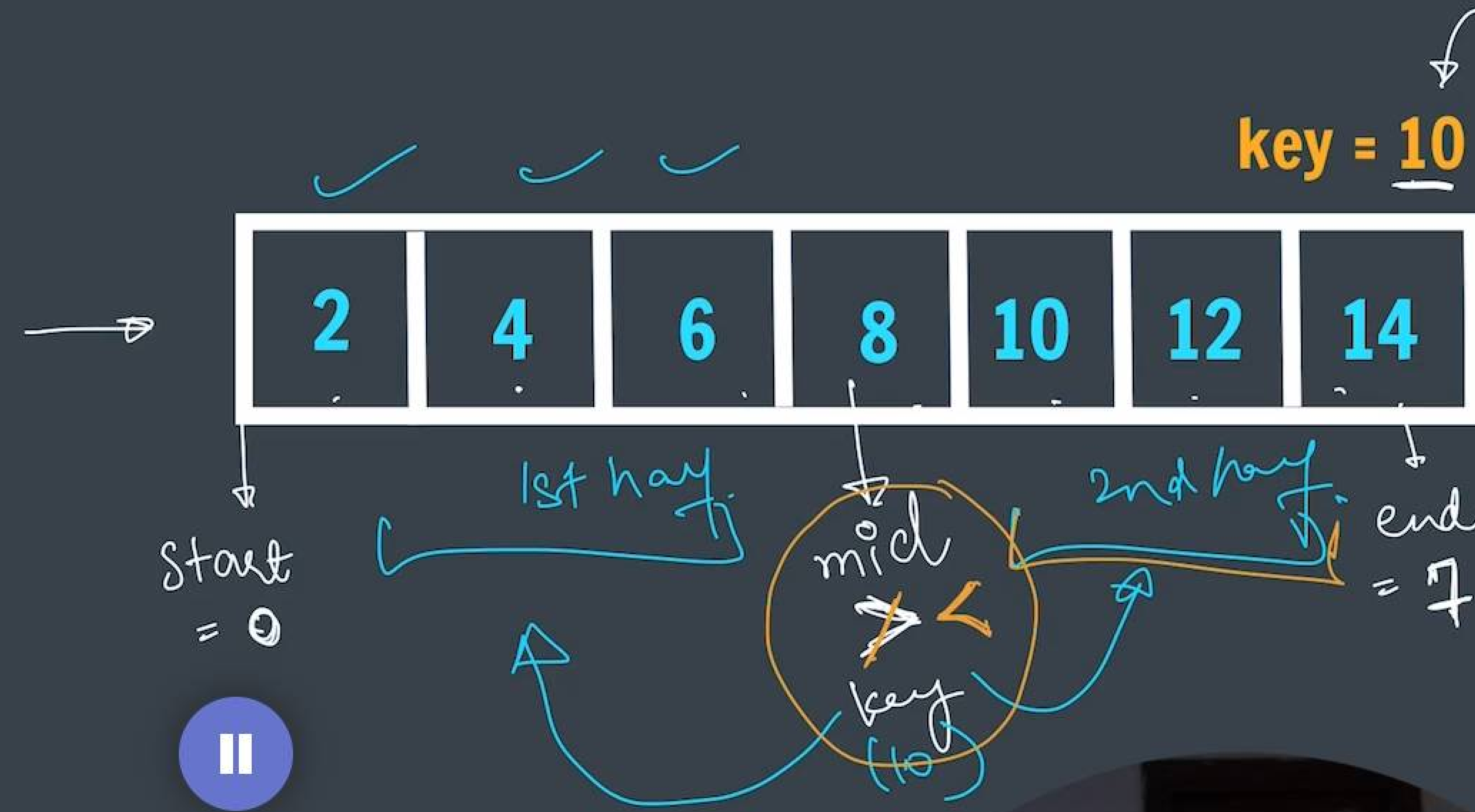
mango

apple banana cheeku litchi mango strawber

ARRAYS

Binary Search

prerequisite - sorted arrays



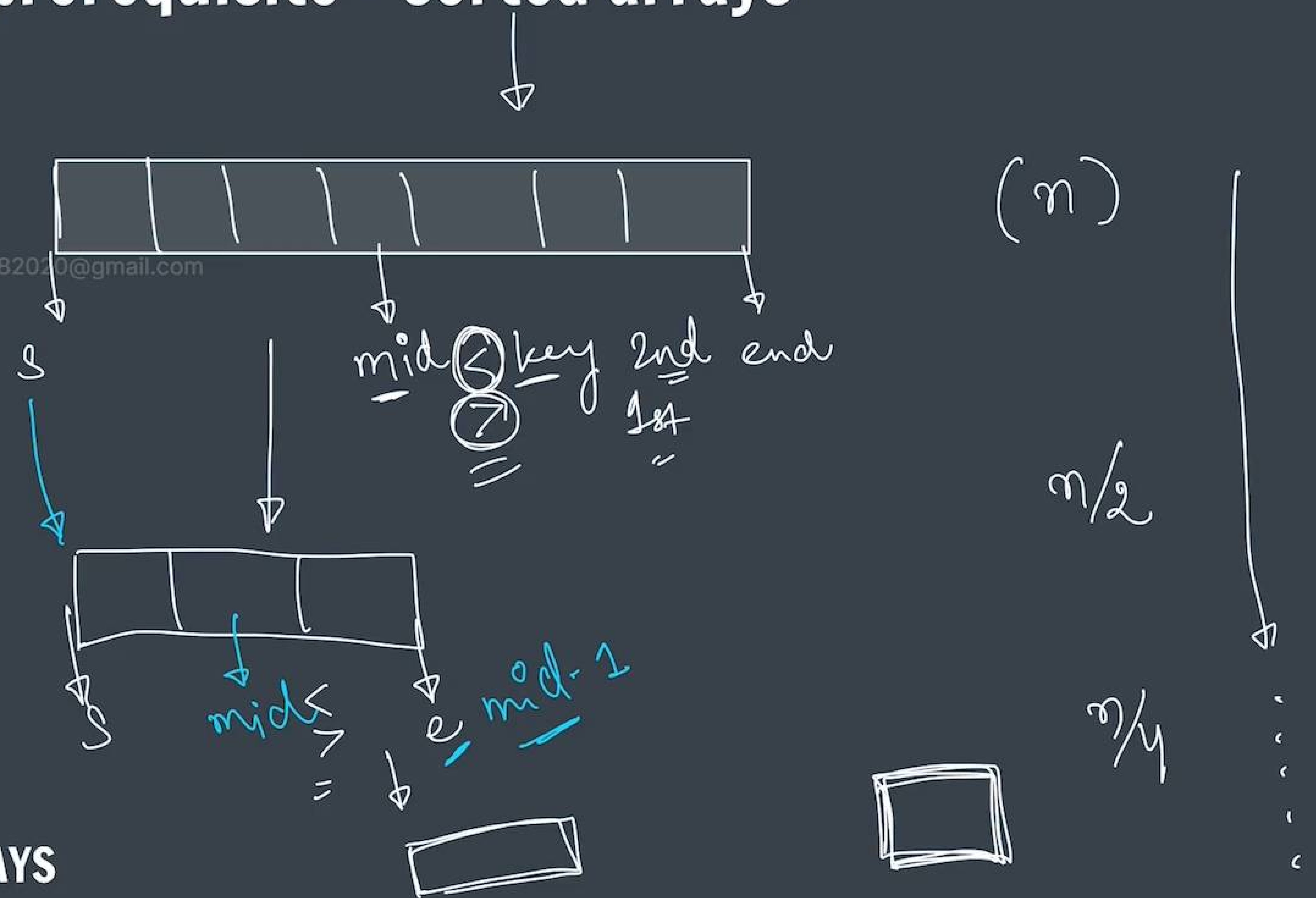
ARRAYS

Binary Search

key = 10

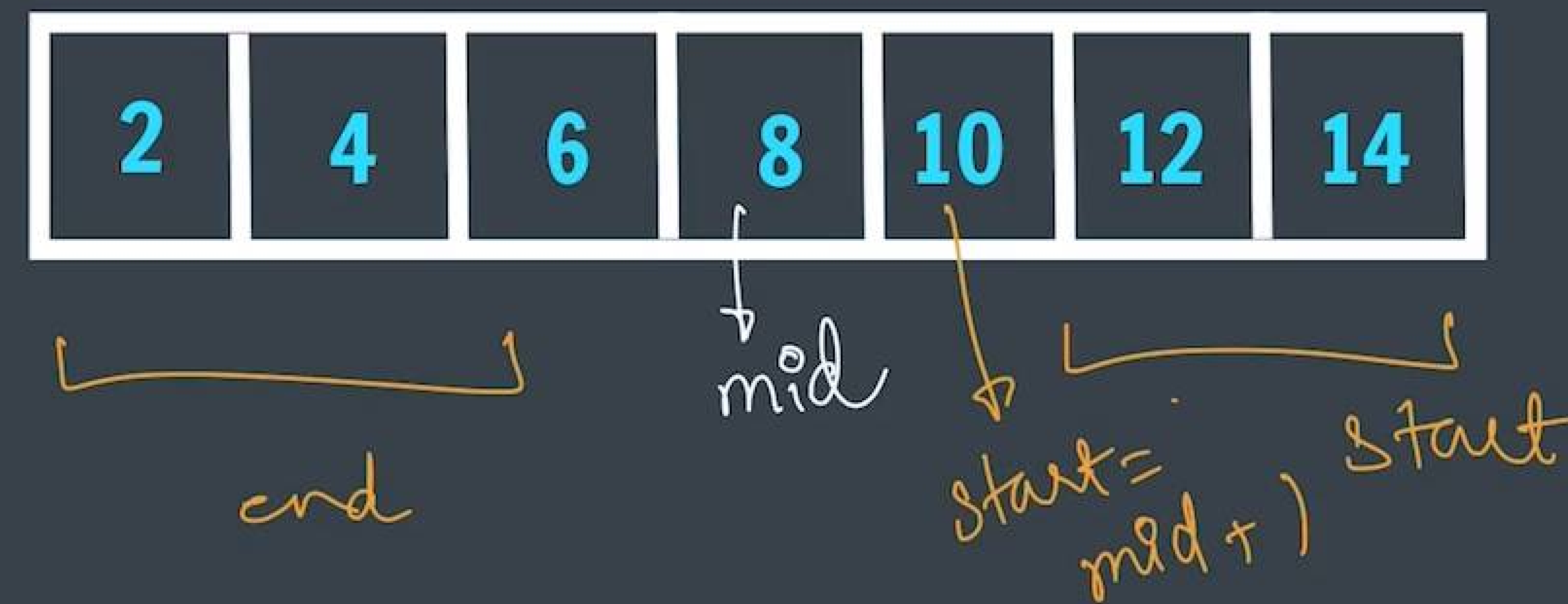
2	4	6	8	10	12	14

prerequisite - sorted arrays



ARRAYS

PseudoCode



start=0, end=n-1

while(start <= end)

find mid $\underbrace{(start + end) / 2}$

compare mid & key

mid == key	FOUND ✓		
mid > key	LEFT	1st	end =
mid < key	RIGHT	2nd	start

ARRAYS

ArraysCC.java 3, U

ArraysCC.java > ArraysCC > binarySearch(int[], int)

```java
public class ArraysCC {
    public static int binarySearch(int numbers[], int key) {
        int start = 0, end = numbers.length-1;

        while(start <= end) {
            int mid = (start + end) / 2;

            //comparisons
            if(numbers[mid] == key) { //found
                return mid;
            }
            if(numbers[mid] < key) { // right
                start = mid+1;
            } else { //left
                end = mid-1;
            }
        }

        return -1;
    }

    Run | Debug
    public static void main(String args[]) {
```


Time Complexity

2	4	6	8	10	12	14

Iteration 1 → n → $n/2^0$

Iteration 2 → $n/2$ → $n/2^1$

Iteration 3 → $n/4$ → $n/2^2$

Iteration 4 → $n/8$ → $n/2^3$

⋮

1 → $n/2^k = 1$ (k+1)th iteration

$$\frac{n}{2^k} = 1 \Rightarrow n = 2^k$$

$$\boxed{k = \log_2 n}$$

$$TC \propto \log_2 n$$

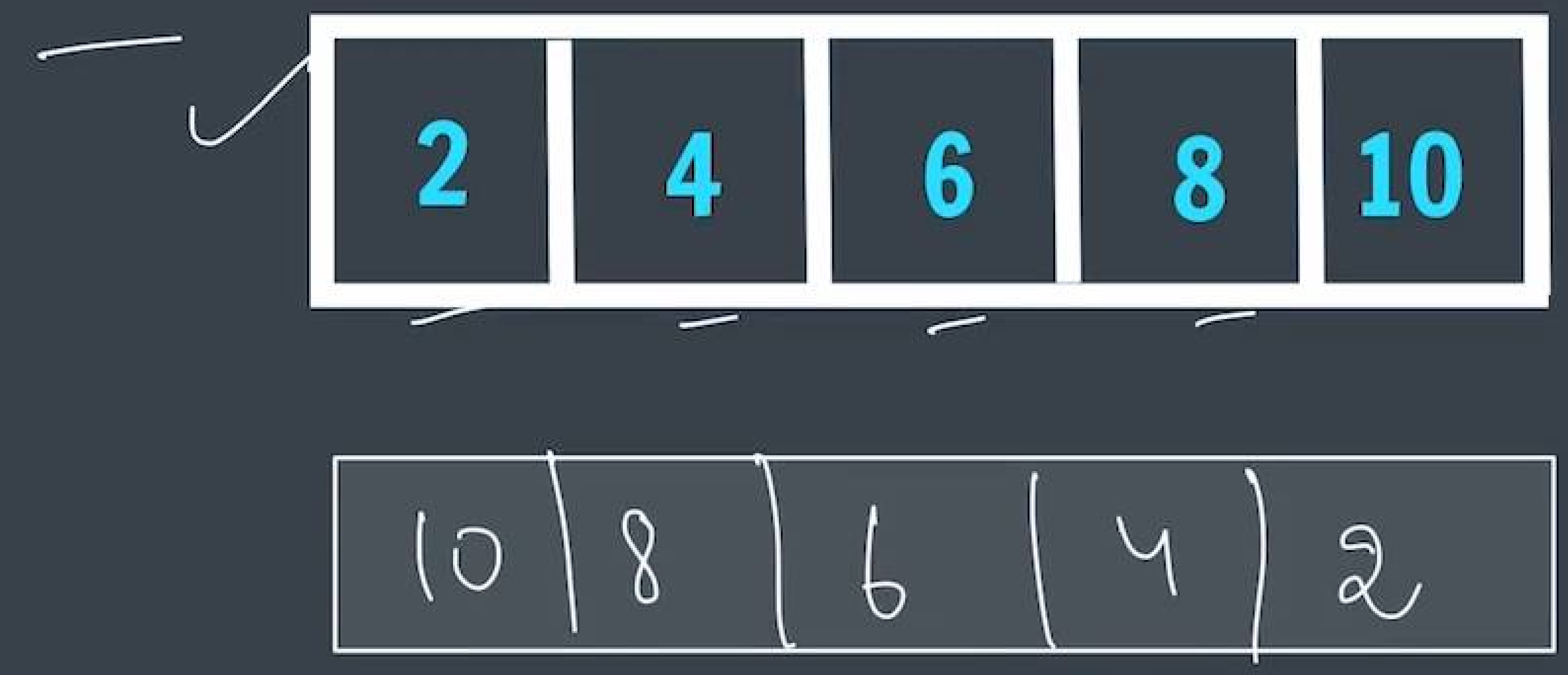
$$O(\log n)$$

$$n > \boxed{\log_2 n}$$

LS BS

ARRAYS

Reverse an Array



ARRAYS

Reverse an Array

2	4	6	8	10

copy

10	8	6	4	2

10	8	6	4	2

space ✗

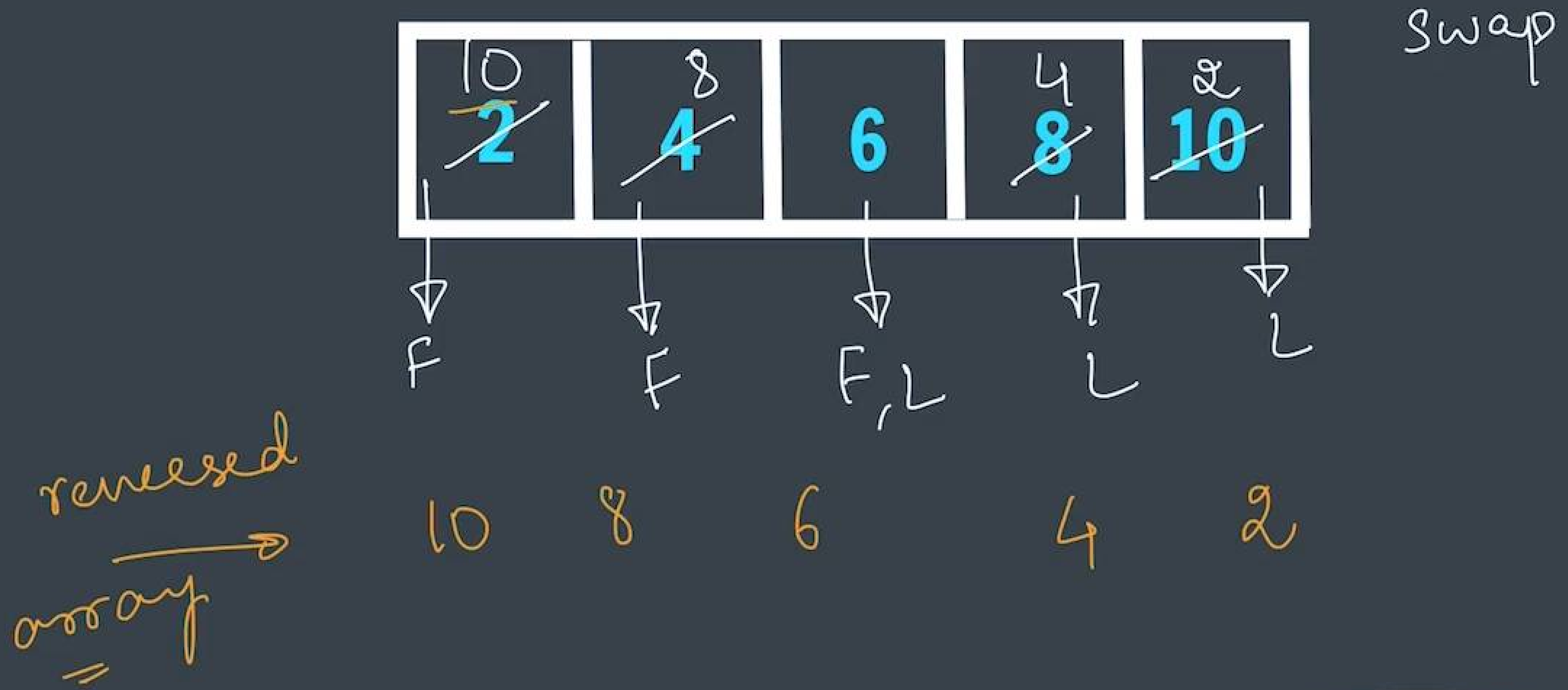
TC = O(n) ✓

SC = ~~O(n)~~

O(1)

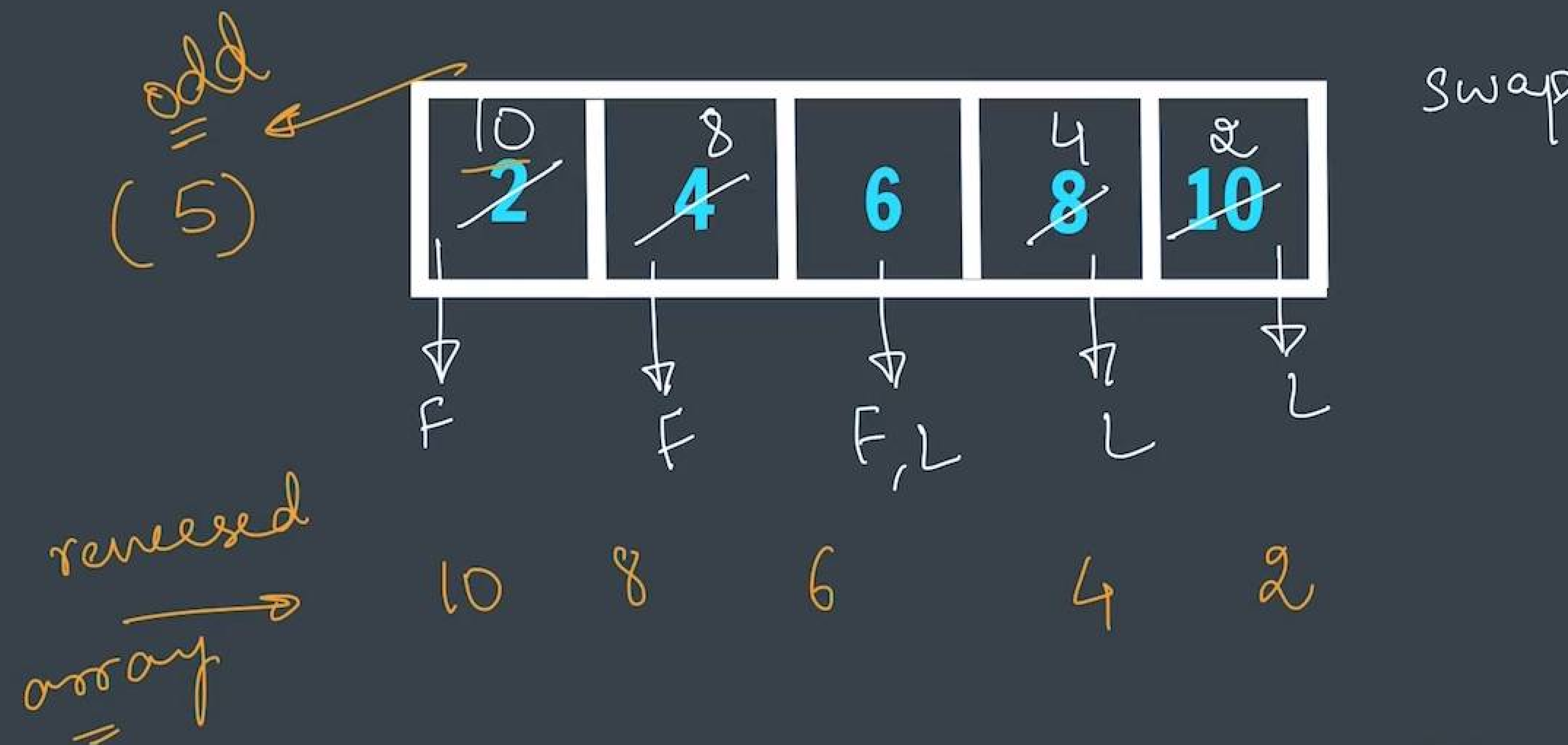
ARRAYS

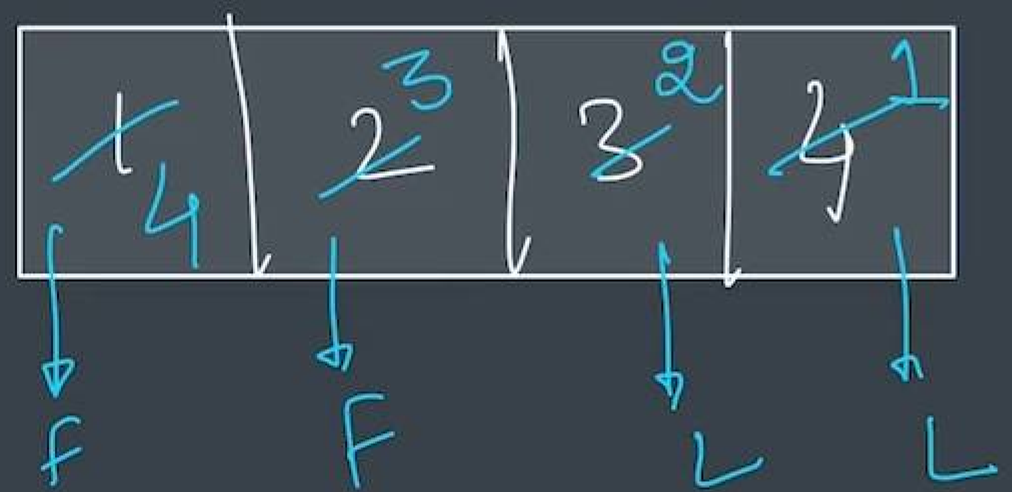
Reverse an Array



ARRAYS

Reverse an Array





4 3 2 1 ← reversed

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ArraysCC.java 2, U

ArraysCC.java > ArraysCC > main(String[])

```java

public class ArraysCC {
    public static void reverse(int numbers[]) {
        int first = 0, last = numbers.length-1;

        while(first < last) {
            //swap
            int temp = numbers[last];
            numbers[last] = numbers[first];
            numbers[first] = temp;

            first++;
            last--;
        }
    }

    Run | Debug
    public static void main
        int numbers[] = {2,

        reverse(numbers);
        for(int i=0; i<numb)
    }
```

```
numbers : int[]
Number - java.lang
NumberFormatException - java.lang
NumberEditor - javax.swing.JSpinner
NumberFormat - java.text
NumberFormatProvider - java.text.sp
NumberFormatter - javax.swing.text
NumberOfDocuments - javax.print.at
NumberOfInterveningJobs - javax.pr
NumberUp - javax.print.attribute.s
NumberUpSupported - javax.print.att
```

ArraysCC.java 1, U

ArraysCC.java > ArraysCC > main(String[])

```java
            last--;
        }
    }

    Run | Debug
    public static void main(String args[]) {
        int numbers[] = {2, 4, 6, 8, 10};

        reverse(numbers);

        //print
        for(int i=0; i<numbers.length; i++) {
            System.out.print(numbers[i]+" ");
        }
        System.out.println();
    }
}
```

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ArraysCC.java > ArraysCC > main(String[])

```java
            last--;
        }
    }

    Run | Debug
    public static void main(String args[]) {
        int numbers[] = {2, 4, 6, 8, 10};

        reverse(numbers);

        //print
        for(int i=0; i<numbers.length; i++) {
```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
10 8 6 4 2
shradhakhapra@Shradhas-Air Classroom Codes %
```

Pairs in an Array

2	4	6	8	10

(2, 4) (2,6) (2, 8) (2,10) → 4

(4, 6) (4,8) (4, 10) → 3

(6, 8) (6,10) → 2

(8, 10) → 1

nested loops

ArraysCC.java 2, U

ArraysCC.java > ArraysCC > printPairs(int[])

```java
import java.util.*;
//Pairs in an Array

public class ArraysCC {
    public static void printPairs(int numbers[]) {
        for(int i=0; i<numbers.length; i++) {
            int curr = numbers[i]; // 2, 4, 6, 8, 10
            for(int j=i+1; j<numbers.length; j++) {
                System.out.print("(" + curr + "," + numbers[j] + ") ");
            }
            System.out.println();
        }
    }

    Run | Debug
    public static void main(String args[]) {
        int numbers[] = {2, 4, 6, 8, 10};



    }
}
```


ArraysCC.java 1, U

ArraysCC.java > ArraysCC > printPairs(int[])

```java
//Pairs in an Array

public class ArraysCC {
    public static void printPairs(int numbers[]) {
        for(int i=0; i<numbers.length; i++) {
            int curr = numbers[i]; // 2, 4, 6, 8, 10
            for(int j=i+1; j<numbers.length; j++) {
                System.out.print("(" + curr + "," + numbers[j] + ") ");
            }
            System.out.println();
        }
    }
```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
(2,4) (2,6) (2,8) (2,10)
(4,6) (4,8) (4,10)
(6,8) (6,10)
(8,10)

shradhakhapra@Shradhas-Air Classroom Codes %
```

```java
        for(int i=0; i<numbers.length; i++) {
            int curr = numbers[i]; // 2, 4, 6, 8, 10
            for(int j=i+1; j<numbers.length; j++) {
                System.out.print("(" + curr + "," + numbers[j] + ") ");
                tp++;
            }
            System.out.println();
        }
        System.out.println("total pairs = " + tp);
    }

```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE zsh

```
(4,6) (4,8) (4,10)
(6,8) (6,10)
(8,10)

shradhakhapra@Shradhas-Air Classroom Codes % javac ArraysCC.java
shradhakhapra@Shradhas-Air Classroom Codes % java ArraysCC.java
(2,4) (2,6) (2,8) (2,10)
(4,6) (4,8) (4,10)
(6,8) (6,10)
(8,10)

total pairs = 10
shradhakhapra@Shradhas-Air Classroom Codes %
```

main* Run Testcases ⓧ 0 ⚠ 1 Spaces: 4 UTF-8 LF Java Go Live

Pairs in an Array

2	4	6	8	10

(2, 4) (2,6) (2, 8) (2,10) (4)

(4, 6) (4,8) (4, 10) (3)

(6, 8) (6,10)

(8, 10)

for (int i = 0 to n)

current

for (int j = i+1 to n) [4, 6, 8, 10]

$\Rightarrow$ pairs

$i = n-1$

2 | 4 | 6 | 8 | 10

n elements

5

$$Tp = \frac{n(n-1)}{2} \qquad \frac{5(4)}{2} = 10$$

$$1 + 2 + 3 + 4 \Rightarrow \boxed{\frac{n(n-1)}{2}}$$

Pairs in an Array

2	4	6	8	10

(2, 4) (2,6) (2, 8) (2,10)

(4, 6) (4,8) (4, 10)

(6, 8) (6,10)

(8, 10)

$O(n^2)$

nested loop

$O(n)$

$O($

$n + (n-1) + (n-2) + (n-3) - - \quad 1$

AP

$\propto O(n^2)$

ARRAYS

Print Subarrays

2	4	6	8	10

n

$\left\{ \frac{n(n+1)}{2} \right\}$

a continuous part of array

2   2, 4   2, 4, 6   2, 4, 6, 8   2, 4, 6, 8, 10 → 5

4   4, 6   4, 6, 8   4, 6, 8, 10 → 4

6   6, 8   6, 8, 10 → 3

8   8, 10 → 2

10 → 1

ARRAYS

Print Subarrays

2	4	6	8	10

nested loops

a **continuous** part of array

start

end

for (int i=0 to n) — L1

for (int j = i+1 to n) — L2

for (start to end) — L3

subarray

ARRAYS

Print Subarrays

2	4	6	8	10

a continuous part of array

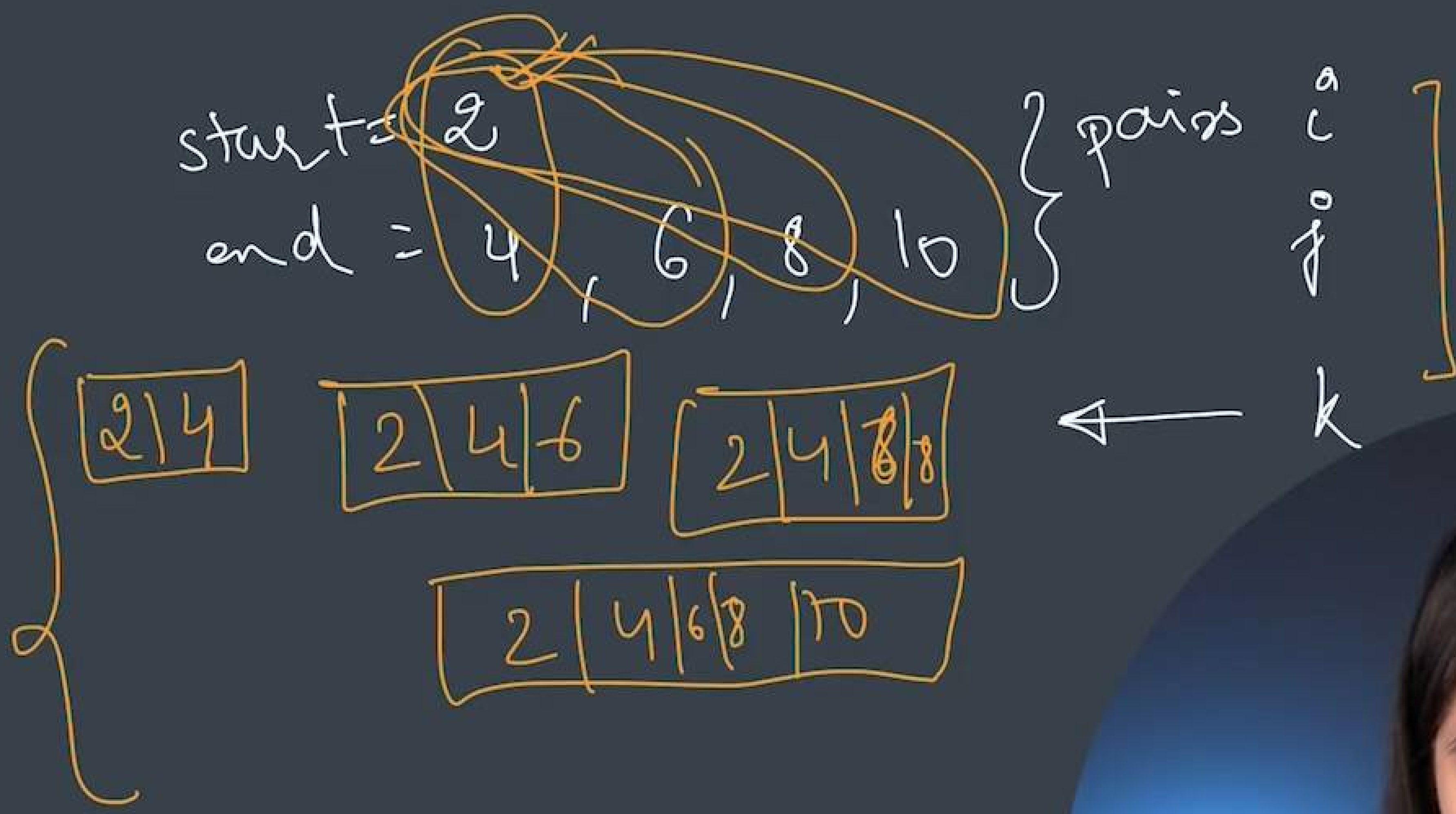
↓ nested loops

start for (int i=0 to n) — L1

end for (int j=i+1 to n) — L2

for (start to end) — L3

{ subarray }



ArraysCC.java 2, U

ArraysCC.java > ArraysCC > printSubarrays(int[])

```java
import java.util.*;
//Print Subarrays

public class ArraysCC {
    public static void printSubarrays(int numbers[]) {
        for(int i=0; i<numbers.length; i++) {
            int start = i;
            for(int j=i; j<numbers.length; j++) {
                int end = j;
                for(int k=start; k<=end; k++) { //print
                    System.out.print(numbers[k]+" "); //subarray
                }
                System.out.println();
            }
            System.out.println();
        }
    }
    public static void main(String args[]) {
        int numbers[] = {2, 4, 6, 8, 10};

    }
}
```

armankhan082020@gmail.com

ArraysCC.java 1, U ×

ArraysCC.java > ArraysCC > main(String[])

```java
import java.util.*;
//Print Subarrays

public class ArraysCC {
    public static void printSubarrays(int numbers[]) {
```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE zsh

```
2
2 4
2 4 6
2 4 6 8
2 4 6 8 10

4
4 6
4 6 8
4 6 8 10

6
6 8
6 8 10

8
8 10

10

shradhakhapra@Shradhas-Air Classroom Codes %
```

n082020@gmail.com

ArraysCC.java 1, U ×

ArraysCC.java > ArraysCC > printSubarrays(int[])

```java
    public static void printSubarrays(int numbers[]) {
        int ts = 0;
        for(int i=0; i<numbers.length; i++) {
            int start = i;
            for(int j=i; j<numbers.length; j++) {
                int end = j;
                for(int k=start; k<=end; k++) { //print
                    System.out.print(numbers[k]+" "); //subarray
                }
                ts++;
                System.out.println();
            }
            System.out.println();
        }

        System.out.println("total subarrays = " + ts);
    }
```

Run | Debug

PROBLEMS 1   OUTPUT   TERMINAL   DEBUG CONSOLE

```
8
8 10

10

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```

Print Subarrays

2	4	6	8	10

a continuous part of array

"2" $\longrightarrow$ 5

4 $\longrightarrow$ 4

6 $\longrightarrow$ 3

8 $\longrightarrow$ 2

10 $\longrightarrow$ 1

$5 + 4 + 3 + 2 + 1 = 15$

$\text{sum } n \longrightarrow \left\{ \frac{n(n+1)}{2} \right\}^{tc}$

ARRAYS